DB-Curator

Agentic database construction for scientific evidence

A tool that searches literature, proposes candidates, and verifies each database row with auditable evidence

Chen Zhao, Botao Liu, Xiaoxiao Li, Tao Huan

The task: build a database, not just answer a question

Scientific pain point

Specialized scientific databases are still mostly built by expert literature curation.

The hard part is not finding chemical names; it is deciding whether each name belongs in the database.

Release requires identity, topic scope and a paper trail that another scientist can inspect.

Agentic framing

The literature set is built during the run: searches change as candidates and evidence gaps appear.

Each candidate becomes an evidence contract: required predicates must pass and exclusion contexts must fail.

The system stops with an explicit state: supported, rejected, or evidence-limited.

What users get

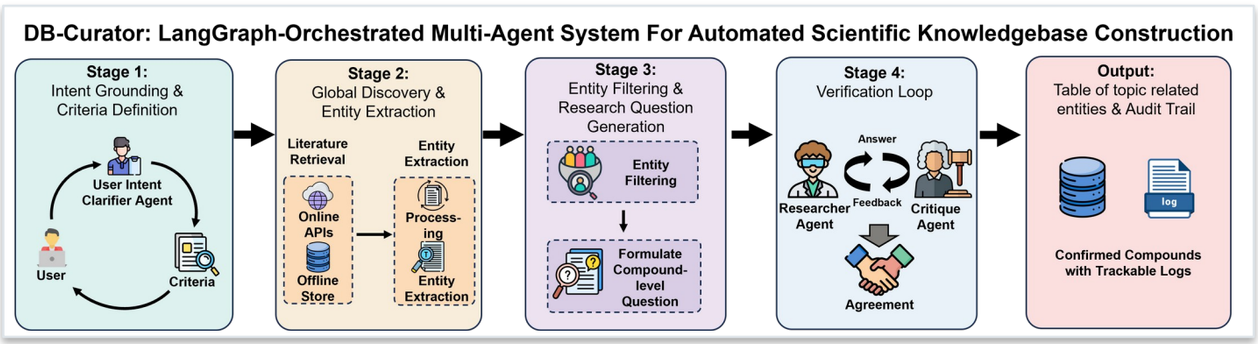
Curated compound rows with identifiers, topic labels and decision states.

Cited papers plus quote or local-context support for the decision.

A compact audit trail so humans review hard cases instead of searching from scratch.

One-sentence story: DB-Curator replaces open-ended manual literature curation with an agentic workflow that searches, verifies and leaves an audit trail.

How the workflow connects search to verification



How to read this figure: left side builds the candidate pool; right side checks whether each candidate satisfies the rubric.

1. Dynamic literature

Broad search first; targeted search follows when a candidate or missing predicate needs evidence.

2. Candidate grounding

Ground chemical names, then ask: does this compound fit the database definition?

3. Predicate verification

Researcher finds evidence; Critique checks whether required predicates are supported.

supported **unresolved**

exclusion **exhausted**

What the system produces: release-ready rows

246/250

release-accepted rows

Overlap plus human-accepted non-overlap rows.

178

benchmark overlaps

Known database entries recovered automatically.

68

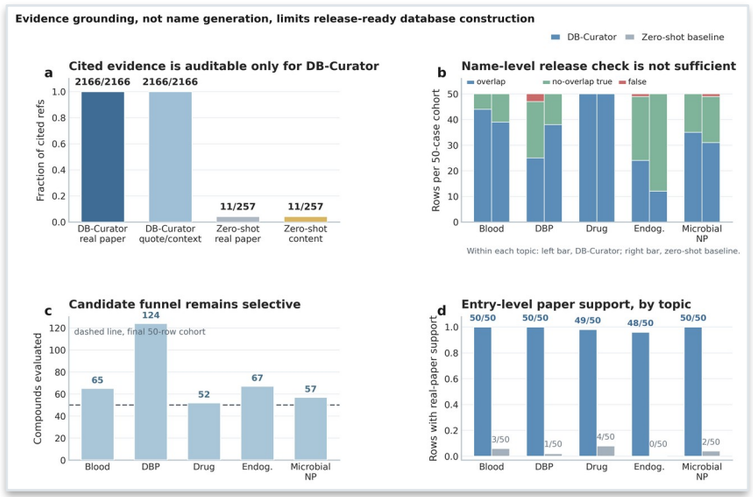
new valid entries

Absent from benchmark, accepted by human experts.

4

errors

False or out-of-scope rows.



How to read the figure: DB-Curator is strong because accepted names also carry auditable paper evidence.

Why this matters scientifically

Benchmarks are useful but incomplete for open-ended database building.

A useful system should recover known compounds and also surface evidence-backed additions.

The human role shifts from searching everything to auditing the few non-overlap rows and their evidence.

How every accepted output is verified

Paper-level audit

First ask: is the cited item a real primary paper? Resolve it using DOI, PMID, title or metadata.

Then ask: does the paper actually contain the quoted span or equivalent local context?

Compound lookup pages are useful for identity, but they are not counted as primary-paper evidence.

Predicate-level audit

A paper must support the actual rubric predicate, not just mention a related chemical or context.

Reagent, solvent and out-of-scope contexts block release even if the compound name is real.

If evidence is missing, the agent searches again or leaves the row evidence-limited.

2,166/2,166

fixed-cohort support

Refs with quote/context support.

7,430/7,453

benchmark-scale support

Auditable refs across five domains.

11/257

zero-shot baseline

Usable refs from direct generation.

Presenter cue: stress that the output is not trusted because the model says so; it is trusted because the cited paper and rubric predicate can be inspected.

Benchmark-scale evidence and agentic correction

1,887

evaluated candidates

Five scientific domains.

977

system Confirmed

Rows judged supported.

7,430/7,453

auditable refs

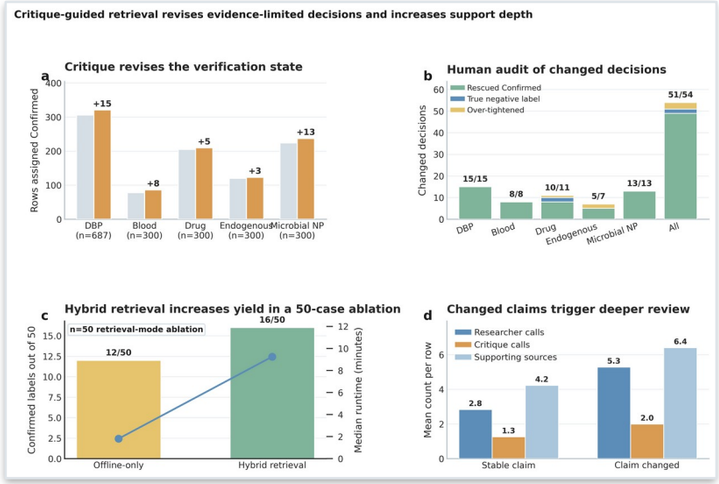
Refs with quote/context support.

49 + 5

critique changes

49 rescued; 5 tightened.





Presenter cue: use this slide to connect scale with mechanism: evidence-grounded decisions, plus critique-driven correction.

Takeaway: a pattern for agentic AI in science

1. Scientific application

DB-Curator turns an open-ended database request into candidate rows.

It recovers known entries and surfaces benchmark-missing, human-validated additions.

Humans audit compact evidence trails instead of searching the literature from scratch.

2. Verification principle

Accepted rows require real papers plus quote or local-context support.

The paper must support the actual predicate, not just mention a nearby concept.

Unresolved cases remain evidence-limited rather than being forced into release.

3. ML lesson

The agentic value is not fluent generation.

It is dynamic evidence construction, explicit state tracking and critique-driven correction.

The output is an auditable artifact, not a one-shot answer.

Final message for the audience: agentic systems become useful in science when they make decisions inspectable, correctable and grounded in evidence.